SIMULATION TASK

```
clc;
clear;
close;

// ========================================
// STATE SPACE MATRICES
// ========================================

A = [0 1; -4 -3];
B = [0; 1];
C = [1 0];
D = 0;

// ========================================
// TIME
// ========================================

dt = 0.01;
t = 0:dt:10;
N = length(t);

// ========================================
// STEP RESPONSE
// ========================================

x = [0; 0];
```

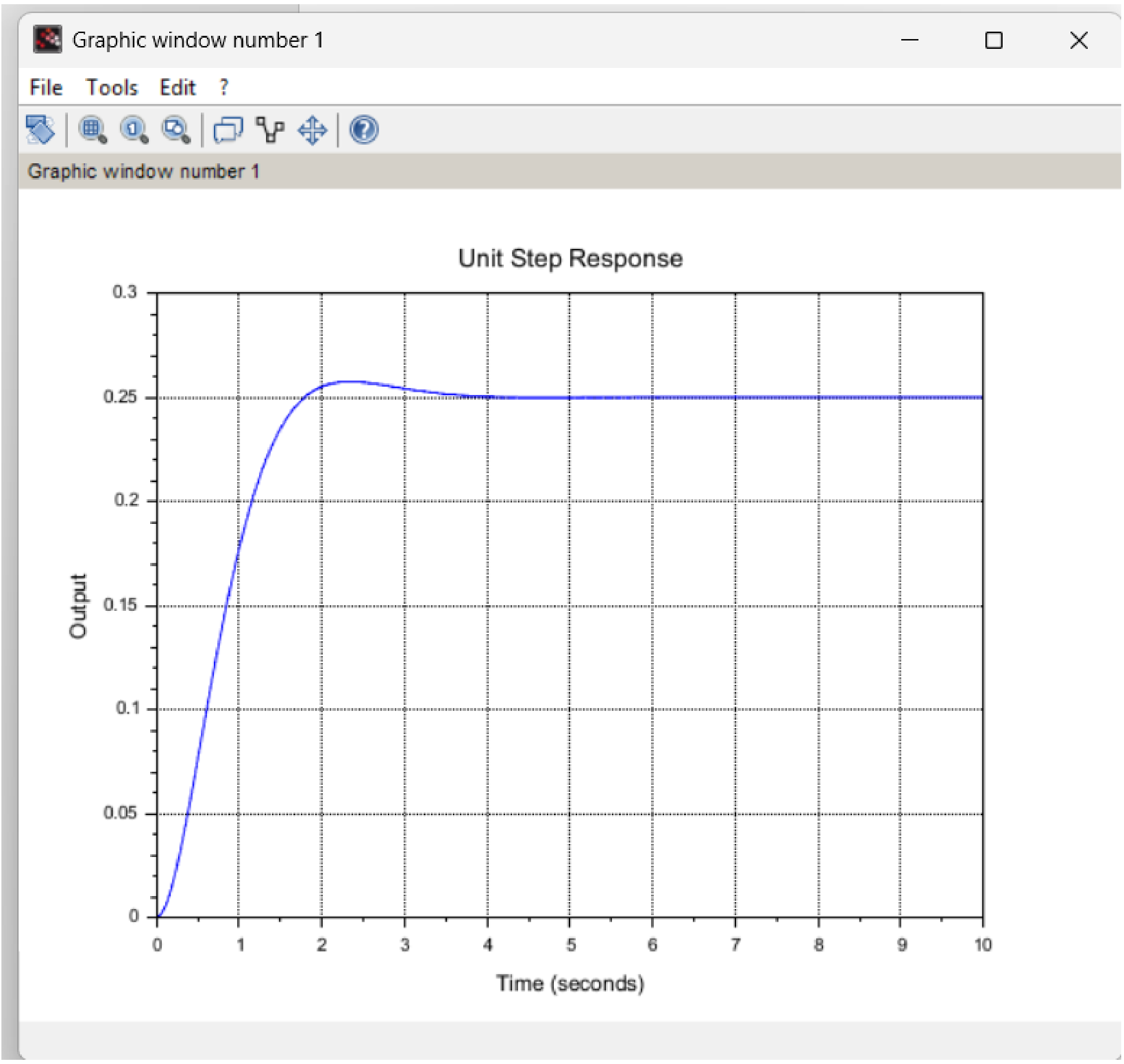
```
y_step = zeros(1,N);

for k = 1:N
    u = 1;

    dx = A*x + B*u;
    x = x + dx*dt;

    y_step(k) = C*x + D*u;
end

// ========================================
// RAMP RESPONSE
// ========================================

x = [0; 0];
y_ramp = zeros(1,N);

for k = 1:N
    u = t(k);

    dx = A*x + B*u;
    x = x + dx*dt;

    y_ramp(k) = C*x + D*u;
end
```

```
// ========================================
// PLOT STEP RESPONSE
// ========================================

scf(1);
plot(t, y_step);
xgrid();

title("Unit Step Response");
xlabel("Time (seconds)");
ylabel("Output");

// ========================================
// PLOT RAMP RESPONSE
// ========================================

scf(2);
plot(t, y_ramp);
xgrid();

title("Ramp Input Response");
xlabel("Time (seconds)");
ylabel("Output");

// ========================================
// EIGENVALUES / POLES
// ========================================
```

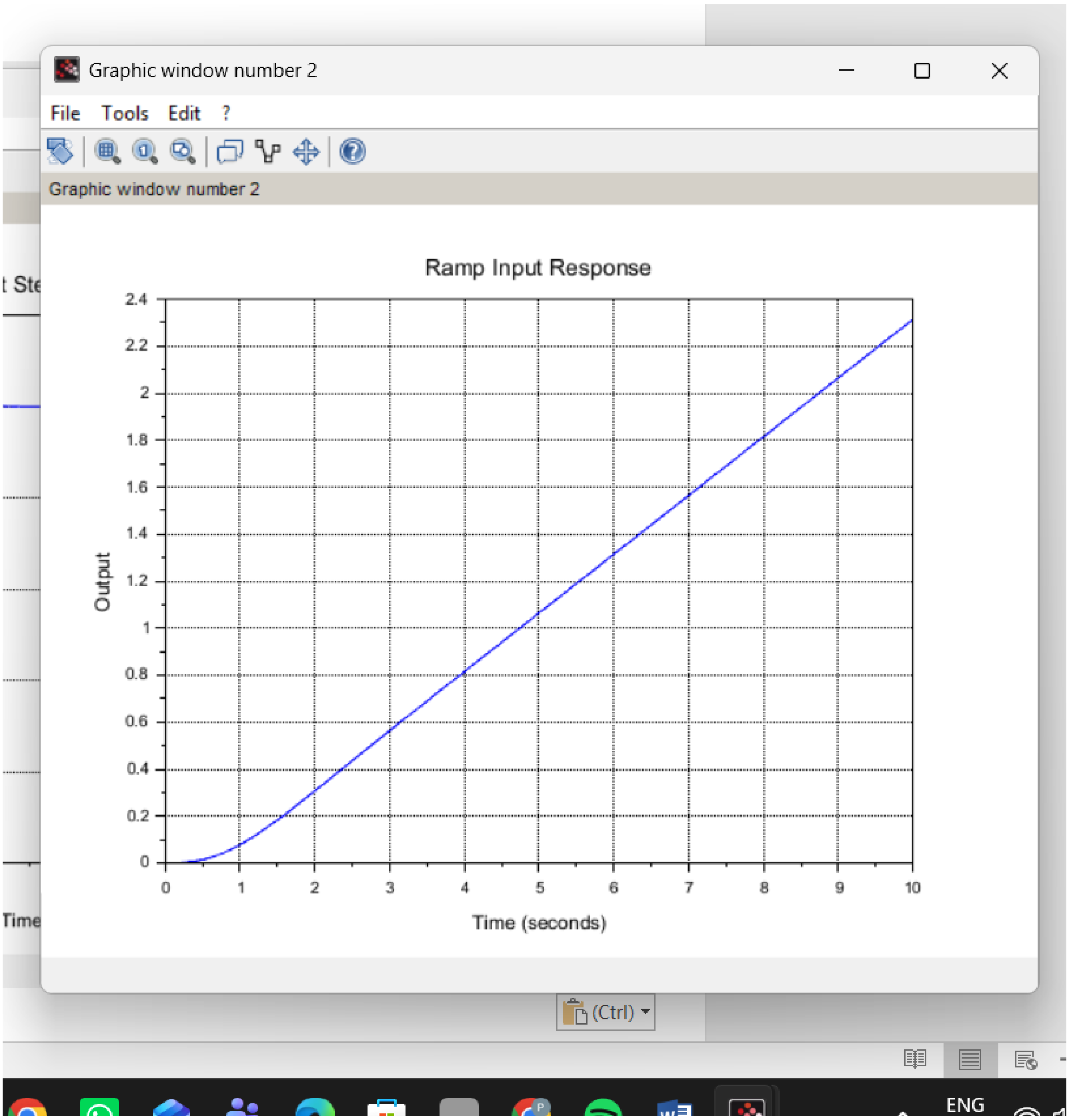
```
p = spec(A);

disp("====================================");
disp("Eigenvalues / Poles:");
disp(p);
disp("====================================");

// ========================================
// STABILITY CHECK
// ========================================

if real(p(1)) < 0 & real(p(2)) < 0 then
    disp("System is STABLE");
else
    disp("System is UNSTABLE");
end

// ========================================
// SETTLING TIME - 2% CRITERION
// ========================================

final_value = y_step(N);

settling_time = 0;

for k = 1:N
```

```
    if abs(y_step(k) - final_value) > 0.02*abs(final_value) then
        settling_time = t(k);
    end
end

disp("=================================");
disp("Final Value:");
disp(final_value);

disp("Settling Time (2%):");
disp(settling_time);

disp("=================================");
```

OUTPUT:



Graphic window number 1
File Tools Edit ?
Graphic window number 1
Unit Step Response
Output
Time (seconds)
0.3
0.25
0.2
0.15
0.1
0.05
0
0 1 2 3 4 5 6 7 8 9 10



Graphic window number 2
File
Tools
Edit
?
Graphic window number 2
Ramp Input Response
Output
Time (seconds)
0
1
2
3
4
5
6
7
8
9
10
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
2.2
2.4
(Ctrl)
ENG